

Oliver Neal

Games Developer · Unity (C#)

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Middlesbrough, UK · Right to work in the UK · Open to relocation and remote work

PROFILE

Games developer specialising in Unity (C#) gameplay systems: state-machine-driven characters and enemy AI, local co-op and multiplayer plumbing, and the profiling work that keeps a scene inside its frame budget. Making games in Unity since 2018, and building them consistently since 2023. First Class Honours graduate in Games Development from Teesside University, and winner of Best Games Programming and Development Artefact at ExpoTees 2026. Currently a gameplay programmer at Shiverbug Studios on *Out of Water*, open to gameplay, systems or AI programming roles from September 2026. Nine shipped projects, most playable in the browser, with source on GitHub and technical breakdowns at oliverneal.dev.

EXPERIENCE

Gameplay Programmer (Internship) · Shiverbug Studios

Jul to Sep 2026

- Own the character and enemy gameplay code on *Out of Water*, an original-IP 3D splitscreen platformer collectathon for two players, working in engine daily.
- Rebuilt both playable characters and the enemies from single-file controllers onto component-driven finite and hierarchical state machines, and built the co-op moveset on top.
- Ran the performance pass: GPU-driven rendering, quality tiers, LOD and culling, measured branch to branch with a profiling tool I wrote for the team.
- Built the splitscreen plumbing (per-player device routing, cameras, character swapping) and a UI Toolkit front end on a shared token system: HUD, character select, pause, save slots and a controls tutorial.
- Implemented a decoupled FMOD audio layer that the audio team plugs into directly, plus blockout interactables and editor tooling for the level designers.
- Authored the water, caustics and bubble shaders, in Shader Graph and in hand-written URP HLSL.
- Designed and shipped the studio's public website, shiverbugstudios.com, alongside gameplay work.

Unity 6 · C# · FSM/HSM · Performance · UI Toolkit · FMOD · Cinemachine · Shader Graph · HLSL

TECHNICAL SKILLS

Languages & engines: C#, Unity 6 (2D/3D), Shader Graph, HLSL, Unreal Engine 5, Python

Gameplay & AI: gameplay programming, systems design, finite & hierarchical state machines, AI navigation (NavMesh), local co-op & splitscreen, Unity Input System

Performance: profiling and frame analysis, time-sliced & multi-rate simulation, GPU-driven rendering, LOD and culling, quality tiers, runtime NavMesh baking

Audio, UI & presentation: FMOD, UI Toolkit, Unity UI, Cinemachine, particle VFX

Multiplayer & services: Netcode for GameObjects, Unity Authentication, Relay, Cloud Save

AI-assisted development: Claude Code and Codex as accelerators in a spec-first workflow, with architecture, implementation direction and code review staying mine

Tools & workflow: Git and Git LFS, branch workflow on a shared team repo, Visual Studio, JetBrains Rider, Figma, Trello, technical documentation

HIGHLIGHTED SOLO PROJECTS

Cyber Station · Unity 3D · C# · AI navigation · simulation Apr 2026

- Neon station management game where passenger AI, train scheduling, staff, economy and a six-factor rating system run as one simulation on a central fixed tick.
- Final-year artefact: scored 90/100 and won Best Games Programming and Development Artefact at ExpoTees 2026.
Breakdown: oliverneal.dev/cyber-station.html · Play: olivernealdev.itch.io/cyber-station

Tile Turfer · Unity 2D · C# · procedural generation · enemy AI Jan 2026

- Turf-war shooter where one territory percentage drives difficulty, fire rate and enemy AI at once, across procedurally generated caves with runtime 2D NavMesh baking.
Breakdown: oliverneal.dev/tile-turfer.html · Play: olivernealdev.itch.io/tile-turfer · Source: github.com/OliverNealDev/TileTurfer

Minimalists · Unity 3D · C# · enemy AI · RTS Aug 2025

- Node-conquest RTS pared back to essentials: capture structures and commit forces by percentage against enemy AI that scales from duels to four-way free-for-alls.
Breakdown: oliverneal.dev/minimalists.html · Play: olivernealdev.itch.io/minimalists · Source: github.com/OliverNealDev/Minimalists

Wanted · Unity 6 · C# · stealth AI · one-week game jam May 2025

- Top-down stealth survival game built in one week for the Tyne to Game jam: a single detection meter drives three-state officer AI, escalating patrols and helicopter passes on a runtime-baked 2D NavMesh.
Breakdown: oliverneal.dev/wanted.html · Play: olivernealdev.itch.io/wanted · Source: github.com/OliverNealDev/Wanted

Unicellular · Unity · C# · FSM · optimisation Feb 2025

- Idle simulation stress-tested at 1,000 and 10,000 concurrent entities, kept smooth by a decoupled multi-rate tick and time-sliced proximity AI.
Breakdown: oliverneal.dev/unicellular.html · Play: olivernealdev.itch.io/unicellular

3D Netcode Tank Game · Unity 3D · C# · netcode · multiplayer Jan 2025

- Server-authoritative multiplayer tank shooter: Relay-hosted sessions, cloud authentication, a decoupled turret aiming rig, and a 64-ray hitscan resolved on the server.
Breakdown: oliverneal.dev/tank-game.html · Source: github.com/OliverNealDev/NetcodeTankGame

Pinnable · Unity 2D · C# · arcade · physics Mar 2024

- Arcade score-attack catch game with a weighted spawn director, homing and drilling pin behaviours, and a physics-driven catch loop.
Breakdown: oliverneal.dev/pinnable.html · Play: olivernealdev.itch.io/pinnable · Source: github.com/OliverNealDev/Pinnable

TEAM PROJECTS

Space Bar Simulator · Unreal Engine 5 · Blueprints · team of 10 Jan 2026

- Retro-futurist first-person bar simulator for Beta Arcade, Teesside's studio-simulation module: owned the player controller on Enhanced Input, a component-based interaction system behind a Blueprint interface, camera-blended stations and the UI computer terminal.
Breakdown: oliverneal.dev/space-bar.html

Candy Corporate · Unity 2D · small team · pixel art Apr 2024

- Top-down extraction game inspired by Lethal Company; led level design, tile creation and pixel-art integration.
Play: jurgenwithwings.itch.io/candycorporate

EDUCATION

BSc (Hons) Games Development · Teesside University Jul 2026

- First Class Honours, 80% overall; final-year project scored 84%, its artefact 90/100.

FdSc Games Technologies · Newcastle College University Centre Jul 2025

- 2:1 (Year 1 Distinction, Year 2 Merit).

BTEC Level 3 National Diploma, Creative Digital Media · Newcastle College 2023

- Distinction.